Chapter 3
Discrete Time Markov Chain: Part II

- Stationary probability
- Transient behavior
- Time reverse
- Branching process
- Markov chain Monte Carlo

Stationary Probability

- The long run proportions π_j, $j \geq 0$, are often called stationary probabilities.
- If the initial state is chosen according to the probabilities π_j, $j \geq 0$, then the probability of being in state j at any time n is also equal to π_j.

Known: $\Pi = (\pi_0, \pi_1, \cdots)$. $\Pi = \Pi P = \cdots = \Pi P^n$

$$P(X_n = j) = \sum_k P(X_n = j \mid X_0 = k) P(X_0 = k)$$

$P^n(k, j)$ $\quad$ π_k

Since $(\pi_0, \pi_1, \cdots) A = (y_0, y_1, \cdots)$, $y_j = \sum_k A(k, j) \pi_k$, then

$P(X_n = j) = y_j$ with $A = P^n$ = the j-entry of $\Pi P^n = \Pi$, which is π_j.

Examples

- **Example 4.27** Suppose the numbers of families that check into a hotel on successive days are independent Poisson random variables with mean λ. Also suppose that the number of days that a family stays in the hotel is a geometric random variable with parameter p, $0 < p < 1$. Also suppose that all families act independently of each other. Under these conditions it is easy to see that if X_n denotes the number of families that are checked in the hotel at the beginning of day n then $\{X_n, n \geq 0\}$ is a Markov chain. Find
  (a) the transition probabilities of this Markov chain;
  (b) $E[X_n|X_0 = i]$;
  (c) the stationary probabilities of this Markov chain.

No lecture on this

Stationary Probability

- **Proposition 4.6** Let $\{X_n, n \geq 1\}$ be an irreducible Markov chain with stationary probabilities π_j, $j \geq 0$, and let r be a bounded function on the state space. Then, with probability 1,

$$\lim_{N\to\infty} \frac{\sum_{n=1}^{N} r(X_n)}{N} = \sum_j r(j)\pi_j.$$

The right hand side $= Er(X)$, where $P(X=j) = \pi_j$ for $j = 0, 1, 2, \cdots$. This is the law of large numbers of discrete markov chain

Proof. Let $a_j(N)$ be the # of times the Mc stay in state j during time periods $1, 2, \cdots, N$. Then

$$\sum_{n=1}^{N} r(X_n) = \sum_{j=0}^{\infty} a_j(N)\, r(j) \quad \text{and} \quad \frac{1}{N}\sum_{n=1}^{N} r(X_n) = \sum_{j=0}^{\infty} \frac{a_j(N)}{N}\, r(j)$$

Since $\frac{a_j(N)}{N} \to \pi_j$, let $N \to \infty$ we get $\frac{1}{N}\sum_{n=1}^{N} r(X_n) \to \sum_{j=0}^{\infty} \pi_j r(j)$.

Non-lecture material

We actually have the following.

Thm. Let X_n be an irreducible MC with stationary dist $\pi_j, j \geq 0$. Let r be a fun s.t. $\sum_j |r(j)| \pi_j < \infty$. Then, w.p. 1,

$$\lim_{N\to\infty} \frac{1}{N} \sum_{n=1}^{N} r(X_n) = \sum_j r(j)\pi_j.$$

The proof is a based on a few lemmas.

Given x, let $X_0 = x$. Define $T_0 = 0$ & $T_{n+1} = \{ i > T_n ; X_i = x \}$, $n = 0, 1, 2, \cdots$. By the def. of stationary property, X_n is positive recurrent, hence $T_n < \infty$ for all $n \geq 1$.

Lemma 1. $\{T_1, T_2 - T_1, T_3 - T_2, \cdots\}$ are iid r.v.'s.

Lemma 2. $X_1, \cdots, X_{T_n}$ and $T_{n+1} - T_n$ are indept for $n \geq 1$.

Proof. $\forall$ fun f and g, we need to show

$$E f(X_1, \cdots, X_{T_n}) g(T_{n+1} - T_n) = E f(X_1, \cdots, X_{T_n}) \cdot E g(T_{n+1} - T_n).$$

In fact,

$$\text{LHS} = E \sum_{\substack{k\geq 1 \\ \ell \geq 1}} f(X_1, \cdots, X_k)\, g(\ell)\, 1(T_n = k,\ T_{n+1} = k+\ell)$$

$$= E \sum_{\substack{k\geq 1 \\ \ell \geq 1}} E\left[f(X_1, \cdots, X_k)\, g(\ell)\, 1(T_n = k,\ T_{n+1} = k+\ell) \,\middle|\, X_1, \cdots, X_k \right]$$

Non-lecture material

$$= E \sum_{\substack{k \geq 1 \\ \ell \geq 1}} f(X_1, \dots, X_k)\, g(\ell)\, 1(T_n = k) \cdot$$

$$E\left[1(T_n = k,\ T_{n+1} = k+\ell) \,\middle|\, X_1, \dots, X_k \right]$$

Write down the last expectation: it is equal to

$$E\, 1(X_{k+\ell} = x,\ X_i \neq x \text{ for } k+1 \leq i < k+\ell,\ X_k = x,\ X_i \neq x \text{ for } 1 \leq i < k$$

$$\mid X_1, \dots, X_k = x)$$

We write $1(X_i \neq x)$ as a sum of $1(X_i = y)$. Then by the MC property, the above is

$$p\left(X_{k+\ell} = x,\ X_i \neq x \text{ for } k+1 \leq i < k+\ell \,\middle|\, X_k = x\right)$$

$$= p(X_\ell = x,\ X_i \neq x \text{ for } 1 \leq i \leq \ell \mid X_0 = x)$$

$$= p(T_1 = \ell).$$

It follows that

$$\text{LHS} = E \sum_{\substack{k \geq 1 \\ \ell \geq 1}} f(X_1, \dots, X_{T_n})\, 1(T_n = k) \cdot g(\ell)\, p(T_1 = \ell)$$

$$= \underbrace{E \sum_{k \geq 1} f(X_1, \dots, X_{T_n})\, 1(T_n = k)}_{E\, f(X_1, \dots, X_{T_n})} \cdot \underbrace{\sum_{\ell \geq 1} g(\ell)\, p(T_1 = \ell)}_{E\, g(T_1)}$$

$$= E f(X_1, \dots, X_{T_n}) \cdot E\, g(T_{n+1} - T_n)$$

by Lemma 1.

Non-lecture material

Lemma 3. For any $n \geq 1$ and fun f, we have

$$\mathbb{E} f(X_{T_n+1}, \cdots, X_{T_{n+1}}) = \mathbb{E} f(X_1, \cdots, X_{T_n}).$$

Proof. Write

$$\mathbb{E} f(X_{T_n+1}, \cdots, X_{T_{n+1}})$$

$$= \sum_{k > \ell \geq 1} \mathbb{E} f(X_{\ell+1}, \cdots, X_k) \, 1(T_n = \ell, T_{n+1} = k)$$

$$= \mathbb{E} \sum_{k > \ell \geq 1} \mathbb{E}\left[f(X_{\ell+1}, \cdots, X_k) \, 1(T_n = \ell, T_{n+1} = k) \mid X_1, \cdots, X_\ell \right]$$

$$= \mathbb{E} \sum_{k > \ell \geq 1} 1(T_n = \ell) \, \mathbb{E}\left[f(X_{\ell+1}, \cdots, X_k) \, 1(X_k = x, \, X_i \neq x \text{ for } \ell+1 \leq i < k \mid X_\ell = x) \right]$$

$$\overset{\text{Stationary}}{=\!=} \mathbb{E} \sum_{k > \ell \geq 1} 1(T_n = \ell) \, \mathbb{E}\left[f(X_1, \cdots, X_{k-\ell}) \, 1(X_{k-\ell} = x, \, X_i \neq x \text{ for } 1 \leq i \leq k-\ell \mid X_0 = x) \right]$$

[where we write $\{X_i \neq x\}$ as a union of $\{X_i = y\}$.]

$$= \mathbb{E} \sum_{k > \ell \geq 1} 1(T_n = \ell) \, \mathbb{E}\left[f(X_1, \cdots, X_{k-\ell}) \, 1(T_1 = k-\ell) \right]$$

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$$= \sum_{k \geq \ell \geq 1} P(T_n = \ell)\, E\left[f(X_1, \cdots, X_{T_1}) \cdot 1(T_1 = k - \ell) \right]$$

$$= \sum_{\substack{\ell \geq 1 \\ i \geq 1}} P(T_n = \ell)\, E\left[f(X_1, \cdots, X_{T_1})\, 1(T_1 = i) \right]$$

$$\overset{\text{Sum over } \ell \text{ first}}{=\!=\!=} \sum_{i \geq 1} E\left[f(X_1, \cdots, X_{T_1})\, 1(T_1 = i) \right]$$

$$= E\, f(X_1, \cdots, X_{T_1}).$$

Lemma 4. $X_1, \cdots, X_{T_n}$ and $X_{T_n+1}, \cdots, X_{T_{n+1}}$ are indept.

Proof. $\forall$ f and g.

$$E f(X_1, \cdots, X_{T_n})\, g(X_{T_n+1}, \cdots, X_{T_{n+1}})$$

$$= E \sum_{\substack{k \geq 1 \\ \ell \geq 1}} f(X_1, \cdots, X_k)\, g(X_{k+1}, \cdots, X_{k+\ell})\, 1(T_n = k,\ T_{n+1} = k+\ell)$$

$$\overset{\text{Conditing on } X_1, \cdots, X_k = x}{=\!=\!=} E \sum_{\substack{k \geq 1 \\ \ell \geq 1}} f(X_1, \cdots, X_k)\, 1(T_n = k)\, E\left[g(X_{k+1}, \cdots, X_{k+\ell})\, 1(T_{n+1} = k+\ell) \mid X_k = x \right] \quad (*)$$

Notice

Non-lecture material

$$E\left[g(X_{k+1},\cdots,X_{k+\ell})1(T_{n+1}=k+\ell)\,\middle|\,X_k=x\right]$$

$$=\sum_{y_i\neq x,\cdots,y_{k+\ell-1}\neq x} g(y_1,\cdots,y_{k+\ell-1},x)\,P(X_{k+\ell}=x,\ X_i=y_i \text{ for } k+1\le i<k+\ell \mid X_k=x)$$

$$=\sum_{y_i\neq x,\cdots,y_{k+\ell-1}\neq x} g(y_1,\cdots,y_{k+\ell-1},x)\cdot P(X_\ell=x,\ X_i=y_i \text{ for } 1\le i<\ell \mid X_0=x)$$

$$=E\left[g(X_1,\cdots,X_{T_1})\cdot 1(T_1=\ell)\right]. \quad (**)$$

Combine (*) and (**) to see

$$Ef(X_1,\cdots,X_{T_n})g(X_{T_n+1},\cdots,X_{T_{n+1}})$$

$$=E\sum_{k\ge 1} f(X_1,\cdots,X_{T_n})1(T_n=k)\cdot\sum_{\ell\ge 1}E\left[g(X_1,\cdots,X_{T_1})\cdot 1(T_1=\ell)\right.$$

$$=\left[Ef(X_1,\cdots,X_{T_n})\right]\cdot Eg(X_1,\cdots,X_{T_1}).$$

$$=\left[Ef(X_1,\cdots,X_{T_n})\right]\cdot Eg(X_{T_n+1},\cdots,X_{T_{n+1}})$$

by Lemma 3.

By induction, we have

Corollary. $\{(X_{T_n+1},\cdots,X_{T_{n+1}});\ n\ge 0\}$ are i.i.d.

Example

- **Example 4.29** For the four state Bonus Malus automobile insurance system specified in Example 4.7, find the average annual premium paid by a policyholder whose yearly number of claims is a Poisson random variable with mean 1/2.

State	Annual Premium	Next state if			
		0 claims	*1 claim*	*2 claims*	*⩾ 3 claims*
1	200	1	2	3	4
2	250	1	3	4	4
3	400	2	4	4	4
4	600	3	4	4	4

Solution. Let $a_k = e^{-1/2}\frac{(1/2)^k}{k!}$. Then transition matrix is

$$\mathbf{P} = \begin{array}{c} \\ 1 \\ 2 \\ 3 \\ 4 \end{array} \begin{array}{c} \begin{array}{cccc} 1 & 2 & 3 & 4 \end{array} \\ \left\| \begin{array}{cccc} a_0 & a_1 & a_2 & 1-a_0-a_1-a_2 \\ a_0 & 0 & a_1 & 1-a_0-a_1 \\ 0 & a_0 & 0 & 1-a_0 \\ 0 & 0 & a_0 & 1-a_0 \end{array} \right\| \end{array}$$

$a_0 = 0.6065$,

$a_1 = 0.3033$, $a_2 = 0.0758$. The transition proba. matrix:

$$P = \begin{pmatrix} 0.6065 & 0.3033 & 0.0758 & 0.1146 \\ 0.6065 & 0 & 0.3033 & 0.0902 \\ 0 & 0.6065 & 0 & 0.3935 \\ 0 & 0 & 0.6065 & 0.3935 \end{pmatrix}$$

let $\Pi = (\Pi_1, \Pi_2, \Pi_3, \Pi_4)$. Solve

$$\begin{cases} \Pi_1 + \Pi_2 + \Pi_3 + \Pi_4 = 1, \\ \Pi = \Pi P. \end{cases}$$

Solve to get

$\Pi_1 = 0.3692$, $\Pi_2 = 0.2395$, $\Pi_3 = 0.2103$,

$\Pi_4 = 0.1809$

By Prop. 4.6 and taking $r(1) = 200$, $r(2) = 250$, $r(3) = 400$, $r(4) = 600$, the average annual premium for a policyholder is

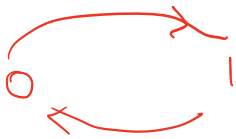
$$\frac{1}{N} \sum_{i=1}^{N} r(X_i) \longrightarrow \sum_{j=1}^{4} r(j)\,\pi(j)$$

$$= 200 \cdot 0.3692 + 250 \cdot 0.2395 + 400 \cdot 0.2103 + 600 \cdot 0.1809 = 326.375$$

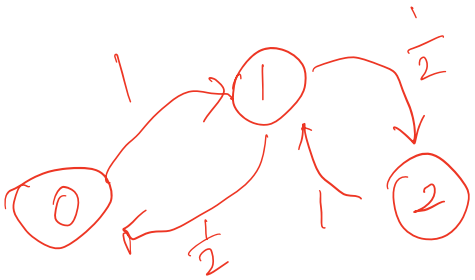
Limiting Probability

- State i is said to have period d if $P_{ii}^n = 0$ whenever n is not divisible by d, and d is the largest integer with this property. A state with period 1 is said to be aperiodic.

$$d = \gcd\{n : P(X_n = i \mid X_0 = i) > 0\}$$



$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $P_{00}^n = \begin{cases} 1, & n \text{ is even} \\ 0, & n \text{ is odd} \end{cases}$, $d = 2$.



$P = \begin{pmatrix} 0 & 1 & 0 \\ \frac{1}{2} & 0 & \frac{1}{2} \\ 0 & 1 & 0 \end{pmatrix}$, $d = 2$

Aperiodic state i: for any $d \geq 2$, there exists $m \geq 1$ s.t. $P_{ii}^{md} = 0$.

- For an irreducible and aperiodic Markov chain, the limiting probabilities will always exist and will not depend on the initial state. Moreover, the limiting probability that the chain will be in state j will equal π_j, the long-run proportion of time the chain is in state j, i.e. for any $i, j \geq 0$,

$$\lim_{n \to \infty} P(X_n = j | X_0 = i) = \pi_j.$$

- An irreducible, positive recurrent, aperiodic Markov chain is said to be ergodic.

Examples

- **Example 4.25 (The Hardy-Weinberg Law and a Markov Chain in Genetics)** Consider a large population of individuals, each of whom possesses a particular pair of genes, of which each individual gene is classified as being of type A or type a. Assume that the proportions of individuals whose gene pairs are AA, aa, or Aa are, respectively, p_0, q_0, and r_0 ($p_0 + q_0 + r_0 = 1$). When two individuals mate, each contributes one of his or her genes, chosen at random, to the resultant offspring. Assuming that the mating occurs at random, in that each individual is equally likely to mate with any other individual, we are interested in determining the proportions of individuals in the next generation whose genes are AA, aa, or Aa.

No lecture on this !

Mean Time Spent in Transient States

- Consider a finite-state Markov chain with transient states $T = \{1, 2, ..., t\}$. Let

$$\mathbf{P}_T = \begin{pmatrix} P_{11} & P_{12} & ... & P_{1t} \\ . & . & . & . \\ . & . & . & . \\ . & . & . & . \\ P_{t1} & P_{t2} & ... & P_{tt} \end{pmatrix}$$

- Suppose s_{ij} is the expected number of times that the Markov chain is in state j given that it starts in state i:

$$s_{ij} = E\left[\sum_{n=0}^{\infty} 1(X_n = j) | X_0 = i\right].$$

Let $\mathbf{S} = (s_{ij})_{1 \le i,j \le t}$, then

$$\mathbf{S} = I + \mathbf{P}_T \mathbf{S} \;\Rightarrow\; \mathbf{S} = (I - \mathbf{P}_T)^{-1}.$$

Proof. Set $\delta_{ij} = 1$ if $i=j$ and $\delta_{ij} = 0$ if $i \neq j$.

Then, for any $i, j \in T$, we have

$$S_{ij} = E\left[\sum_{n=0}^{\infty} 1(X_n = j) \mid X_0 = i\right]$$

$$= \delta_{ij} + E\left[\sum_{n=1}^{\infty} 1(X_n = j) \mid X_0 = i\right]$$

$$= \delta_{ij} + \sum_k E\left[\sum_{n=1}^{\infty} 1(X_n = j) \mid X_0 = i, X_1 = k\right] \cdot \underbrace{P(X_1 = k \mid X_0 = i)}_{P_{ik}}$$

Note that

$$E\left[\sum_{n=1}^{\infty} 1(X_n = j) \mid X_0 = i, X_1 = k\right]$$

$$= E\left[\sum_{n=1}^{\infty} 1(X_n = j) \mid X_1 = k\right]$$

$$= E\left[\sum_{n=0}^{\infty} 1(X_n = j) \mid X_0 = k\right] = S_{kj}$$

But $S_{kj} = 0$ if k is recurrent, since a recurrent state can't go to a transient state

Thus $S_{ij} = \delta_{ij} + \underbrace{\sum_k P_{ik} S_{kj}}_{\to (i,j)\text{-entry of } P_T S}$

$\therefore \quad S = I + P_T S$

$\Rightarrow \quad S = (I - P_T)^{-1}$

$S_{kj} = 0$ if $k \notin T$ since the MC can't go from recurrent to transiente states

Example

- **Example 4.32** Consider the gambler's ruin problem with $p = 0.4$ and $N = 7$. Starting with 3 units, determine
  (a) the expected amount of time the gambler has 5 units,
  (b) the expected amount of time the gambler has 2 units.

Solution. $T = \{1, 2, 3, 4, 5, 6\}$

w.r.t.

$$\mathbf{P}_T = \begin{array}{c|cccccc} & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline 1 & 0 & 0.4 & 0 & 0 & 0 & 0 \\ 2 & 0.6 & 0 & 0.4 & 0 & 0 & 0 \\ 3 & 0 & 0.6 & 0 & 0.4 & 0 & 0 \\ 4 & 0 & 0 & 0.6 & 0 & 0.4 & 0 \\ 5 & 0 & 0 & 0 & 0.6 & 0 & 0.4 \\ 6 & 0 & 0 & 0 & 0 & 0.6 & 0 \end{array}$$

inverting $\mathbf{I} - \mathbf{P}$ gives

We now solve

$$\left(I - P_T\right)^{-1} = \begin{bmatrix} 1.6149 & 1.0248 & 0.6314 & 0.3691 & 0.1943 & 0.0777 \\ 1.5372 & 2.5619 & 1.5784 & 0.9228 & 0.4857 & 0.1943 \\ 1.4206 & 2.3677 & 2.9990 & 1.7533 & 0.9228 & 0.3691 \\ 1.2458 & 2.0763 & 2.6299 & 2.9990 & 1.5784 & 0.6314 \\ 0.9835 & 1.6391 & 2.0763 & 2.3677 & 2.5619 & 1.0248 \\ 0.5901 & 0.9835 & 1.2458 & 1.4206 & 1.5372 & 1.6149 \end{bmatrix}$$

(a) $S_{35} = 0.9228$

(b) $S_{32} = 2.3677$

4.5.1 Gambler's ruin problem

Story: $P(\text{win \$1}) = p$. $P(\text{lose \$1}) = q$

The gambler has wealth $i \in \{0, 1, 2, \cdots, N\}$.

X_n = # of dollars at time n.

Then X_n, $n = 0, 1, 2, \cdots$ is a MC with

$P_{i,i+1} = p$, $P_{i,i-1} = 1-p = q$.

$P_{00} = P_{NN} = 1$.

Let $P_i = P(X_n = N \text{ for some } n \geq 1 \mid X_0 = i)$

$i = 0, 1, \cdots, N$.

Purpose: derive P_i.

Special case: $P_0 = 0$ and $P_N = 1$.

1st, let's look at the 1st gamble, for $1 \leq i \leq N-1$,

$P_i = P(X_n = N \text{ for some } n \geq 1 \mid X_1 = i+1) \cdot P(X_1 = i+1 \mid X_0 = i)$

$+ \quad - \quad - \quad - \quad - \quad \cdot = i-1) \cdot P(X_1 = i-1 \mid X_0 = i)$

$= p \cdot P_{i+1} + q \cdot P_{i-1}$.

Then

$$P_{i+1} - P_i = \frac{q}{p}(P_i - P_{i-1}), \quad 1 \le i \le N-1$$

Then $P_N - P_{N-1} = \frac{q}{p}(P_{N-1} - P_{N-2})$

$$= \left(\frac{q}{p}\right)^2 (P_{N-2} - P_{N-3})$$

$$\vdots$$

$$= \left(\frac{q}{p}\right)^{N-1} (P_1 - P_0)$$

$$= \left(\frac{q}{p}\right)^{N-1} P_1$$

Similarly,

$$P_{i+1} - P_i = \left(\frac{q}{p}\right)^i P_1$$

$i = 1, 2, \cdots, N-1.$

Add all terms up to get

$$P_{i+1} - P_1 = P_1 \left[\left(\frac{q}{p}\right) + \cdots + \left(\frac{q}{p}\right)^i \right]$$

$$\therefore P_{i+1} = \begin{cases} \dfrac{\left(\frac{q}{p}\right)^{i+1} - 1}{\frac{q}{p} - 1} P_1, & p \neq q; \\ (i+1) P_1, & p = q \end{cases}$$

$i = 1, 2, \cdots, N.$

Use $P_N = 1$ and take $i = N-1$ to see

$$P_1 = \begin{cases} \dfrac{\frac{q}{p} - 1}{(\frac{q}{p})^N - 1}, & p \neq q; \\ \dfrac{1}{N}, & p = q. \end{cases}$$

Thus,

$$P_i = \begin{cases} \dfrac{(\frac{q}{p})^i - 1}{(\frac{q}{p})^N - 1}, & \text{if } p \neq q; \\ \dfrac{i}{N}, & \text{if } p = q. \end{cases}$$

Let Q_i be the probab. that gambler loses everything.

Then $Q_i = 1 - P_i$. As a result,

$$Q_i = \begin{cases} \dfrac{(\frac{p}{q})^{N-i} - 1}{(\frac{p}{q})^N - 1}, & \text{if } p \neq q; \\ \dfrac{N-i}{N}, & \text{if } p = q. \end{cases}$$

(Another way to show this: swap "i" & "$N-i$", "$p \to q$").

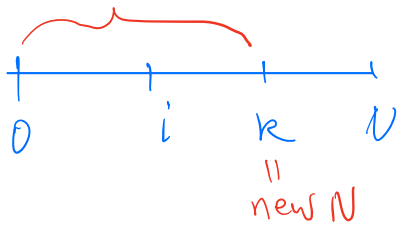
Suppose $0 \leq j < i < k \leq N$. Let

P_{im}

$= P(\text{the gambler's wealth has ever reached \$}m \mid \text{starts at \$}i)$.

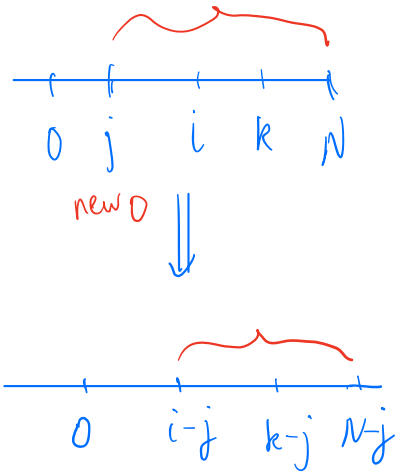
By resetting the absorbing states (changing "N" to "k" & "0" to "j"),

$$P_{ik} = \begin{cases} \dfrac{\left(\frac{q}{p}\right)^i - 1}{\left(\frac{q}{p}\right)^k - 1}, & p \neq q \\ \dfrac{i}{k}, & p = q \end{cases}$$



and

$$P_{ij} = \begin{cases} \dfrac{\left(\frac{p}{q}\right)^{i-j} - 1}{\left(\frac{p}{q}\right)^{N-j} - 1}, & p \neq q; \\ \dfrac{N-i}{N-j}, & p = q. \end{cases}$$



Time Reversible Markov Chain

- A stationary Markov chain observed backwards in time is also Markovian, i.e.

$$P(X_m = j | X_{m+1} = i, X_{m+2}, X_{m+3}, ...) = P(X_m = j | X_{m+1} = i).$$

In this case, the reversed chain has transition probabilities

$$Q_{ij} \triangleq P(X_m = j | X_{m+1} = i) = \frac{\pi_j P_{ji}}{\pi_i}.$$

If $Q_{ij} = P_{ij}$ for all i, j, we say the Markov chain is time reversible.

- **Proposition:** For a time-reversible chain, its stationary probabilities must satisfy

$$\pi_i P_{ij} = \pi_j P_{ji}.$$

On the other hand, if a Markov chain of transition probabilities P_{ij} and there exists $\pi_i \geq 0$ such that

$$\pi_i P_{ij} = \pi_j P_{ji}, \quad \sum_i \pi_i = 1.$$

Then, the chain is time reversible and π_i's are its stationary probabilities.

Proof. Sum $\pi_i P_{ij} = \pi_j P_{ji}$ over $j \geq 0$. Then $\pi_i = \sum_j \pi_j P_{ji} \Rightarrow \pi = \pi P$.

plus $\sum_j \pi_j = 1$, we know the MC is stationary with stationary probab. π.

Examples

- **Example 4.37** Consider a random walk with states 0, 1, . . . , M and transition probabilities

$$\begin{aligned} P_{i,i+1} &= \alpha_i = 1 - P_{i,i-1}, i = 1, 2, ..., M-1, \\ P_{0,1} &= \alpha_0 = 1 - P_{0,0} \\ P_{M,M} &= \alpha_M = 1 - P_{M,M-1}. \end{aligned}$$

- If a positive recurrent Markov chain can only make transitions from a state to one of its two nearest neighbors, it is time reversible.

- **Example 4.37** Consider a random walk with states 0, 1, . . . , M and transition probabilities

$$P_{i,i+1} = \alpha_i = 1 - P_{i,i-1}, i = 1, 2, ..., M-1,$$
$$P_{0,1} = \alpha_0 = 1 - P_{0,0}$$
$$P_{M,M} = \alpha_M = 1 - P_{M,M-1}.$$

Solution. By the Proposition, we only need to find π_i s.t. $\pi_i P_{ij} = \pi_j P_{ji}$ for all i, j and $\sum_i \pi_i = 1$.

Observe $P_{ij} = 0$ if $j \neq i-1$ or $i+1$. Then

$$\pi_0 P_{01} = \pi_1 P_{10} \Rightarrow \pi_1 = \frac{\alpha_0}{1-\alpha_1}\pi_0$$

$$\pi_1 P_{12} = \pi_2 P_{21} \Rightarrow \pi_2 = \frac{\alpha_1}{1-\alpha_2}\pi_1$$

$$\vdots$$

$$\pi_{M-2} P_{M-2,M-1} = \pi_{M-1} P_{M-1,M-2} \Rightarrow \pi_{M-1} = \frac{\alpha_{M-2}}{1-\alpha_{M-1}}\pi_{M-2}$$

$$\pi_{M-1} P_{M-1,M} = \pi_M P_{M,M-1} \Rightarrow \pi_M = \frac{\alpha_{M-1}}{1-\alpha_M}\pi_{M-1}$$

Thus,

$$\pi_1 = \frac{\alpha_0}{1-\alpha_1}\pi_0$$

$$\pi_2 = \frac{\alpha_0\alpha_1}{(1-\alpha_1)(1-\alpha_2)}\pi_0$$

$$\vdots$$

$$\pi_{M-1} = \frac{\alpha_0\alpha_1\cdots\alpha_{M-2}}{(1-\alpha_1)(1-\alpha_2)\cdots(1-\alpha_{M-1})}\pi_0$$

$$\pi_M = \frac{\alpha_0\alpha_1\cdots\alpha_{M-1}}{(1-\alpha_1)(1-\alpha_2)\cdots(1-\alpha_M)}\pi_0$$

Thus

$$1 = \pi_0 + \pi_1 + \cdots + \pi_M$$

$$= \left[1 + \sum_{j=1}^{M} \frac{\alpha_0 \cdots \alpha_{j-1}}{(1-\alpha_1)\cdots(1-\alpha_j)} \right] \pi_0$$

$$\therefore \quad \pi_0 = \left[1 + \sum_{j=1}^{M} \frac{\alpha_0 \cdots \alpha_{j-1}}{(1-\alpha_1)\cdots(1-\alpha_j)} \right]^{-1}.$$

We obtain

$$\pi_i = \frac{\alpha_0 \cdots \alpha_{i-1}}{(1-\alpha_1)\cdots(1-\alpha_i)} \pi_0, \quad i = 1, 2, \cdots, M$$

- In particular, if $\alpha_i \equiv \alpha$, then

$$\pi_0 = \left[1 + \frac{\alpha}{1-\alpha} + \cdots \left(\frac{\alpha}{1-\alpha} \right)^M \right]^{-1} = \left[\frac{1 - \left(\frac{\alpha}{1-\alpha} \right)^{M+1}}{1 - \frac{\alpha}{1-\alpha}} \right]^{-1}$$

$$= \frac{1-\beta}{1-\beta^{M+1}}$$

with $\beta = \frac{\alpha}{1-\alpha}$. $\pi_i = \frac{\alpha^i}{(1-\alpha)^i} \cdot \pi_0 = \frac{(1-\beta)\beta^i}{1-\beta^{M+1}}$

- P. and T. Ehrenfest model (1907)
  - study 2nd law of thermodynamics
  - Cont. version: stochastic diff. eq.



Paul Ehrenfest (1880–1933)



↑

Tatyana Ehrenfest (1905-1984) & Albert Einstein

$$\alpha_i = \frac{M-i}{M}, \quad i = 0, 1, \cdots, M.$$

In this case

$$1 + \sum_{j=1}^{M} \frac{\alpha_0 \cdots \alpha_{j-1}}{(1-\alpha_1)\cdots(1-\alpha_j)} = 1 + \sum_{j=1}^{M} \frac{M \cdot (M-1) \cdots (M-j+1)}{1 \cdot 2 \cdots j}$$

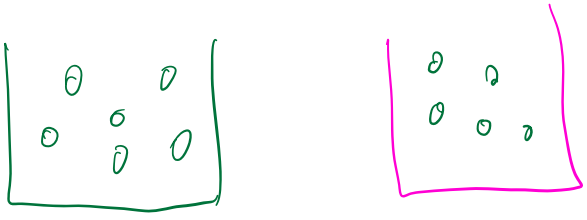
$$= \sum_{j=0}^{M} \binom{M}{j}$$

$$= 2^M.$$

Hence

$$\pi_i = \frac{1}{2^M} \binom{M}{i}, \quad i = 0, 1, 2, \cdots, M.$$

Physical meaning.



Balls labeled $\{1, 2, \cdots, M\}$. Randomly pick a number, get the ball, take out from the urn and place it into the other urn.

X_i = # of balls in the left urn.

$$P_{i,i+1} = \frac{N-i}{N}, \quad P_{i,i-1} = \frac{i}{N}.$$

Our results shows, in the long run, the number of balls in the left urn = # of Heads when flip a fair coin n times indeptly

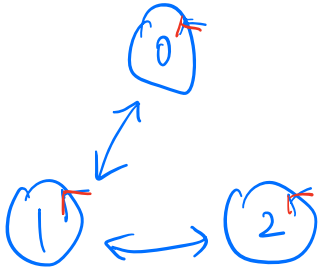
Examples

- **Example 4.10 revisited** An urn always contains 2 balls. Ball colors are red and blue. At each stage a ball is randomly chosen and then replaced by a new ball, which with probability 0.8 is the same color, and with probability 0.2 is the opposite color, as the ball it replaces. What is the long-run probability that a selected ball is red.

Is this chain reversible?

Solution: X_n = # of red in the urn after nth operation.

$$P = \begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array} \begin{array}{c} \begin{array}{ccc} 0 & 1 & 2 \end{array} \\ \begin{pmatrix} 0.8 & 0.2 & 0 \\ 0.1 & 0.8 & 0.1 \\ 0 & 0.2 & 0.8 \end{pmatrix} \end{array}$$



Recurrent, aperiodic.

Solve $\pi_0 + \pi_1 + \pi_2 = 1$

$$(\pi_0\ \pi_1\ \pi_2) = (\pi_0\ \pi_1\ \pi_2)\begin{pmatrix} 0.8 & 0.2 & 0 \\ 0.1 & 0.8 & 0.1 \\ 0 & 0.2 & 0.8 \end{pmatrix}.$$

We get $\pi_0 = \frac{1}{4}, \pi_1 = \frac{1}{2}, \pi_2 = \frac{1}{4}$.

Check: $\pi_i P_{ij} = \pi_j P_{ji}$ for all $i \neq j$.

(1) $\therefore$ The chain is reversible

(2) The chain is irreducible, aperiodic and positive recurrent. Thus, the chain is ergodic. Thus, $\forall$ any i,

$$\lim_{n\to\infty} P(X_n = j \mid X_0 = i) = \begin{cases} \frac{1}{4}, & j = 0 \text{ or } 2; \\ \frac{1}{2}, & j = 1. \end{cases}$$

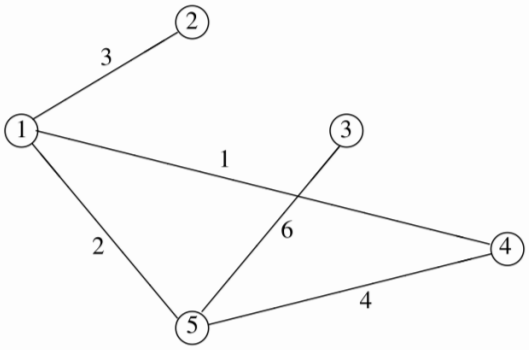
Recall

- For an irreducible and aperiodic Markov chain, the limiting probabilities will always exist and will not depend on the initial state. Moreover, the limiting probability that the chain will be in state j will equal π_j, the long-run proportion of time the chain is in state j, i.e. for any $i, j \geq 0$,

$$\lim_{n\to\infty} P(X_n = j | X_0 = i) = \pi_j.$$

- An irreducible, positive recurrent, aperiodic Markov chain is said to be ergodic.

Example 4.38. Consider an arbitrary connected graph (see Section 3.6 for definitions) having a number w_{ij} associated with arc (i, j) for each arc. One instance of such a graph is given by Fig. 4.1. Now consider a particle moving from node to node in this



- Weighted graph
- Erdos - Reni random graph
- Random walk on graphs
- regular graph $w_{ij} = 1$
- Complex analysis on graphs, diff. Geometry

manner: If at any time the particle resides at node i, then it will next move to node j with probability P_{ij} where

$$P_{ij} = \frac{w_{ij}}{\sum_j w_{ij}}$$

and where w_{ij} is 0 if (i, j) is not an arc. For instance, for the graph of Fig. 4.1, $P_{12} = 3/(3+1+2) = \frac{1}{2}$.

Q: Under what condition, the chain is reversible?

Solution. $\pi_i P_{ij} = \pi_j P_{ji}$ requires

$$\pi_i \frac{w_{ij}}{\sum_j w_{ij}} = \pi_j \frac{w_{ji}}{\sum_i w_{ji}}$$

Since is indirected graph, then $w_{ij} = w_{ji}$. It follows that

$$\frac{\pi_i}{\sum_j w_{ij}} = \frac{\pi_j}{\sum_i w_{ji}}, \quad i \neq j.$$

Let $\dfrac{\pi_i}{\sum_j w_{ij}} = c$ for all i.

Then $1 = \sum_i \pi_i \Longrightarrow c \sum_{i,j} w_{ij} = 1$.

Thus $c = \left(\sum_{i,j} w_{ij}\right)^{-1}$.

We then have

$$\pi_i = \frac{\sum_j w_{ij}}{\sum_{i,j} w_{ij}}.$$

In the picture, $w_{12} = 3$, $w_{14} = 1$, $w_{15} = 2$

$w_{35} = 6$, $w_{45} = 4$

$$\therefore \sum_{j=1}^{5} \sum_{i=1}^{5} w_{ij} = 2(3+1+2+6+4) = 32$$

$$\therefore \pi_1 = \frac{6}{32}, \quad \pi_2 = \frac{3}{32}, \quad \pi_3 = \frac{6}{32}$$

$$\pi_4 = \frac{5}{32}, \quad \pi_5 = \frac{12}{32}.$$

Examples

- **Proposition:** Consider an irreducible Markov chain with transition probabilities P_{ij}. If we can find positive numbers π_i, $i \geq 0$, summing to one, and a transition probability matrix $Q = [Q_{ij}]$ such that

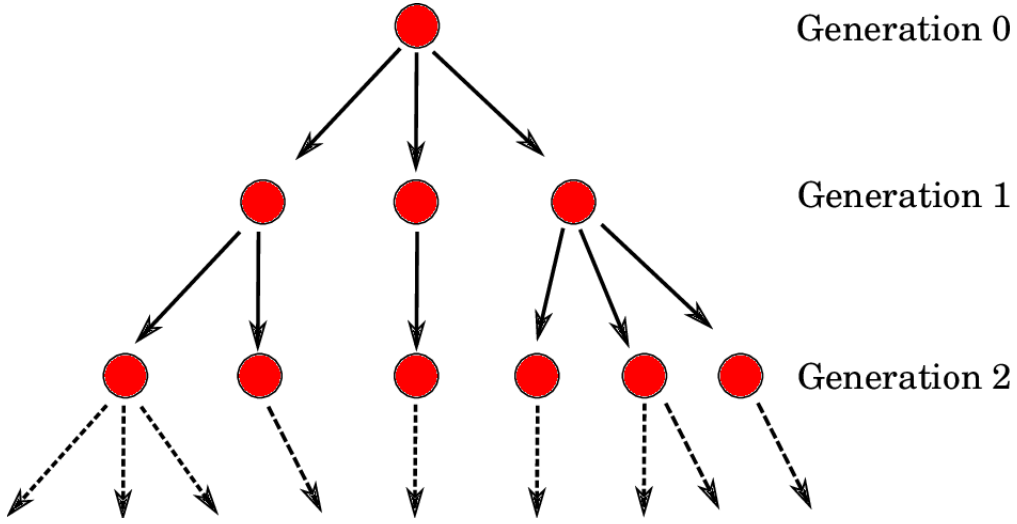
$$\pi_i P_{ij} = \pi_j Q_{ji},$$

  then the Q_{ij} are the transition probabilities of the reversed chain and the π_i are the stationary probabilities both for the original and reversed chain.

- **Example 4.38** A single bulb is necessary to light a given room. When the bulb in use fails, it is replaced by a new one at the beginning of the next day. Let X_n equal i if the bulb in use at the beginning of day n is in its ith day of use (that is, if its present age is i). For instance, if a bulb fails on day $n-1$, then a new bulb will be put in use at the beginning of day n and so $X_n = 1$. If we suppose that each bulb, independently, fails on its ith day of use with probability p_i, $i \geq 1$, then it is easy to see that $\{X_n, n \geq 1\}$ is a Markov chain.

Branching Process

- Branching processes have a wide variety of applications in the biological, sociological, and engineering sciences (including financial engineering).



Model: • $X_0 = 1$, i.e., generation 0 has 1 offspring

• At any generation, each offspring produces its own offsprings with number j according to probab. P_j, $j = 0, 1, 2, \cdots$

Let X_n = # of offsprings at nth generation, Then

$$X_n = \sum_{i=1}^{X_{n-1}} Z_{n,i}, \quad n = 1, 2, \cdots$$

where $Z_{n,i}$ for all i and n are iid r.v.'s with dist

0	1	2	3	4	$\cdots$
P_0	P_1	P_2	P_3	P_4	$\cdots$

We assume $Z_{n,i}$'s are indept of X_{n-1}.

we next calculate EX_n, $Var(X_n)$, dies out probab. $\pi_0 = \lim_{n\to\infty} P(X_n = 0 \mid X_0 = 1)$.

1st, $E(X_n) = E\left(E\left(\sum_{i=1}^{X_{n-1}} Z_{n,i} \mid X_{n-1}\right)\right)$

$= E(X_{n-1}\mu) = \mu E X_{n-1}$

where $\mu = \sum_{j=0}^{\infty} j p_j$ is the mean,

$\sigma^2 = \sum_{j=0}^{\infty} (j-\mu)^2 p_j$ is the variance

Thus by induction

$$\mathbb{E} X_n = \mu \mathbb{E} X_{n-1} = \mu^2 \mathbb{E} X_{n-2} = \mu^3 \mathbb{E} X_{n-3}$$
$$= \cdots = \mu^n \mathbb{E} X_0 = \mu^n.$$

For variance, we have

$$\mathrm{Var}(X_n) = \mathbb{E}\,\mathrm{var}(X_n | X_{n-1}) + \mathrm{Var}(\mathbb{E}(X_n | X_{n-1})).$$

As before, $\mathbb{E}(X_n | X_{n-1}) = \mu X_{n-1}$.

$$\mathrm{Var}(X_n | X_{n-1}) = \mathrm{Var}\left(\sum_{i=1}^{X_{n-1}} Z_{n,i} \,\middle|\, X_{n-1}\right)$$
$$= X_{n-1} \cdot \sigma^2.$$

Thus

$$\mathrm{Var}(X_n) = \sigma^2 \cdot \mathbb{E} X_{n-1} + \mu^2 \cdot \mathrm{Var}(X_{n-1})$$
$$= \sigma^2 \cdot \mu^{n-1} + \mu^2 \cdot \mathrm{Var}(X_{n-1})$$

Thus, $\mathrm{Var}(X_n) = \sigma^2\mu^{n-1} + \mu^2 \cdot \mathrm{Var}(X_{n-1})$

$$= \sigma^2\mu^{n-1} + \mu^2\left(\sigma^2\mu^{n-2} + \mu^2\,\mathrm{Var}(X_{n-2})\right)$$

$$= \sigma^2\left(\mu^{n-1} + \mu^n\right) + \mu^4 \cdot \mathrm{Var}(X_{n-2})$$

$$\therefore \quad \mathrm{Var}(X_n) = \sigma^2\left(\mu^{n-1} + \mu^n\right) + \mu^4\left(\sigma^2\mu^{n-3} + \mu^2 \cdot \mathrm{Var}(X_{n-3})\right)$$

$$= \sigma^2\left(\mu^{n-1} + \mu^n + \mu^{n+1}\right) + \mu^{2\cdot 3}\,\mathrm{Var}(X_{n-3})$$

$$\vdots$$

$$= \sigma^2\left(\mu^{n-1} + \mu^n + \cdots + \mu^{2(n-1)}\right) + \mu^{2n}\,\mathrm{Var}(X_0)$$

$$= \begin{cases} \sigma^2\mu^{n-1}\dfrac{\mu^n - 1}{\mu - 1}, & \mu \neq 1; \\ n\sigma^2, & \mu = 1. \end{cases}$$

Now we calculate $\pi_0 = \lim_{n\to\infty} P(X_n = 0 \mid X_0 = 1)$.

1st, if $\mu < 1$, then

$$\mu^n = EX_n = \sum_{j=1}^{\infty} j\,P(X_n = j)$$

$$\geq P(X_n \geq 1) \quad \text{since } j \geq 1$$

Let $n \to \infty$, $\lim_n P(X_n \geq 1) = 0 \Longrightarrow \pi_0 = 1$.

We also know $\pi_0 = 1$ if $\mu = 1$.

Now assume $\mu > 1$. Then

$$\pi_0 = P(\text{population dies out})$$

$$= \sum_{j=0}^{\infty} P(\text{population dies out} \mid X_1 = j)\, P_j$$

$$= \sum_{j=0}^{\infty} \pi_0^j P_j$$

Since each offspring creates its own offsprings indeptly from others. As a consequence

$$P(\text{population dies out} \mid X_1 = j) = \pi_0^j.$$

Thus $\pi_0 = \sum_{j=0}^{\infty} \pi_0^j P_j$

We can also show

π_0 is the smallest solution of the eq.

Eg. $P_0 = \frac{1}{2}$, $P_1 = \frac{1}{4}$, $P_2 = \frac{1}{4}$

$\mu = \frac{1}{2}\cdot 0 + \frac{1}{4}\cdot 1 + \frac{1}{4}\cdot 2 = \frac{3}{4} < 1$. So $\pi_0 = 1$.

Eg. $p_0 = \frac{1}{4}, p_1 = \frac{1}{4}, p_2 = \frac{1}{2}$

$$\mu = 0 \cdot \frac{1}{4} + 1 \cdot \frac{1}{4} + 2 \cdot \frac{1}{2} = \frac{5}{4} > 1.$$

π_0 satisfies

$$\pi_0 = \frac{1}{4} + \frac{1}{4}\pi_0 + \frac{1}{2}\pi_0^2$$

$$\Rightarrow 2\pi_0^2 - 3\pi_0 + 1 = 0 \Longrightarrow \text{root} : 1, \frac{1}{2}$$

π_0 = Smallest root $= \frac{1}{2}$.

Eg. What if $X_0 = m$ in the previous examples?

Simple! Since each offspring creates its own family indepthly, the die out probab. $= \pi_0^m$.

Therefore, in the 1st example, die out chance $= 1$,

... 2nd ... $\frac{1}{2^m}$.

Now we show the die out prob.

$\pi_0 = P(\text{population dies out})$

is the smallest solution of eq.

$$\pi_0 = \sum_{j=0}^{\infty} \pi_0^j p_j .$$

In fact, define $\varphi(t) = \left(\sum_{j=0}^{\infty} p_j t^j\right) - t$. Obviously

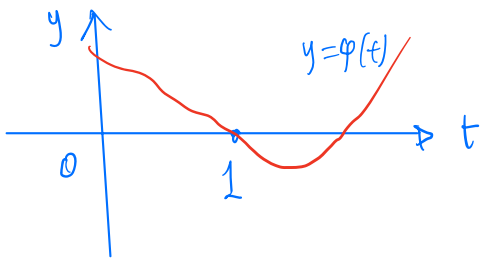
$\varphi(1) = 1$, $\varphi(0) = p_0 > 0$ and

$$\varphi'(t) = \sum_{j=1}^{\infty} j p_j t^{j-1} - 1, \quad \varphi'(1) = \mu - 1,$$

$$\varphi''(t) \geq 0, \quad t \geq 0, \qquad \varphi(t) \text{ is convex.}$$

Thus, $\varphi(t) = 0$ has at most two solutions.

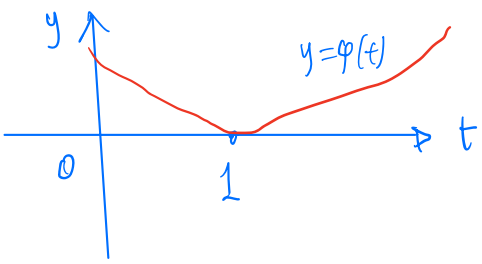
① $\mu \in (0,1)$. Then $\varphi'(1) < 0$. By convexity $\varphi'(x) < 0$ for $x \in [0,1]$.



Thus, $\varphi(x) > \varphi(1) = 0$ for $x \in [0,1)$

There is only one sol in $[0,1]$.

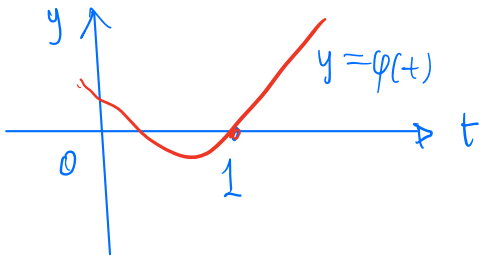
② $\mu = 1$. Then $\varphi'(1) = \varphi(1) = 0$. Easy to see the graph



The sol of $t = \sum_{j=0}^{\infty} t^j p_j$.

is unique : $t = 1$.

(3) $\mu > 1$.



Now $\varphi(1)=0$.

$\varphi'(1) > 0$.

$\Rightarrow \varphi(1-\varepsilon) < 0$ for some $\varepsilon > 0$.

With $\varphi(0) > 0$, we know there is $\pi_0 \in (0,1)$ s.t. $\varphi(\pi_0) = 0$.

Obviously, π_0 and 1 are the only two solutions of $\varphi(z)$. We need to rule out that 1 is not the die-out probab. To do so, we will show

$$\liminf_{n\to\infty} P(X_n = \infty) > 0. \qquad (*)$$

This will imply $\liminf_{n\to\infty} P(X_n = 0) < 1$.

Recall

- $E(X_n \mid X_{n-1}) = \mu X_{n-1}$,
- $E X_n = \mu^n$,
- $\mathrm{Var}(X_n) = \sigma^2 \mu^{n-1} \dfrac{\mu^n - 1}{\mu - 1}$, $\mu > 1$.

The last shows

$$E(X_n^2) = \mu^{2n} + \sigma^2 \mu^{n-1} \frac{\mu^n - 1}{\mu - 1}.$$

Let $Y_n = \frac{X_n}{\mu^n}$. Then Y_n is a martingale. There is a r.v. $Y_\infty \geq 0$ s.t $Y_n \to Y_\infty$ a.s. Also

$$E(Y_n^2) \leq 1 + \frac{\sigma^2}{\mu(\mu-1)}, \quad \forall n \geq 1.$$

This implies $\{Y_n ; n \geq 1\}$ is uniformly integrable. Thus, $1 = EY_n \to EY_\infty$. Hence $EY_\infty = 1$. Thus $P(Y_\infty > 0) > 0$

Thus, on the set $\{Y_\infty > 0\}$, we know $X_n/\mu^n \to Y_\infty$, hence $\lim_{n\to\infty} Y_n = \infty$, we then obtain (*).



Application to Covid 19 Study

- Use branching process to model the transmission of covid.

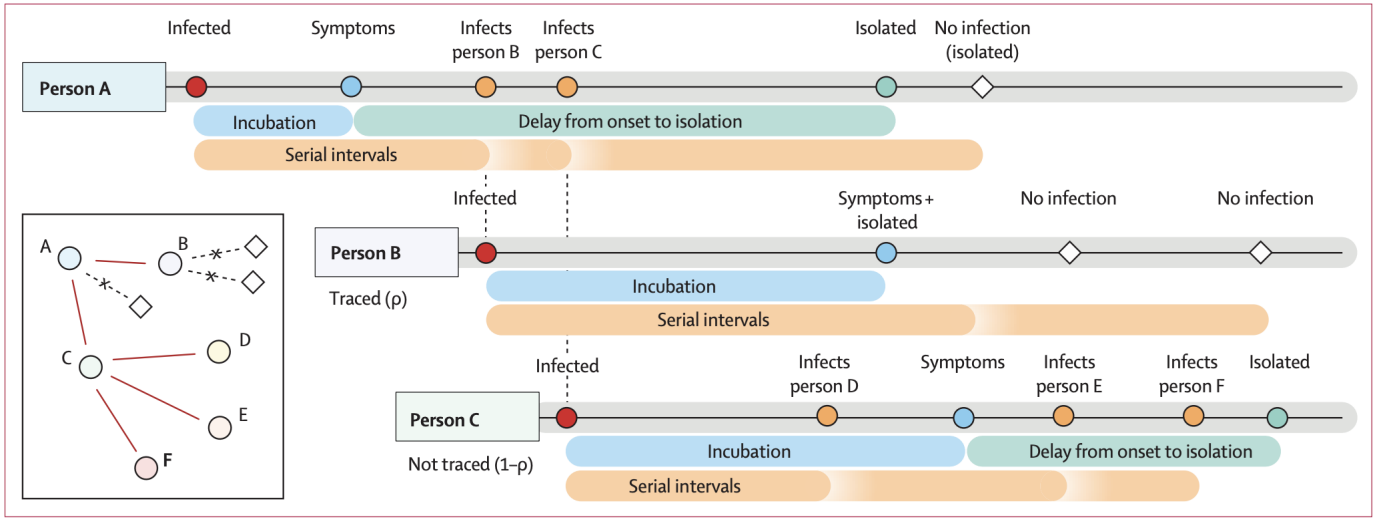


Figure 1: **Example of the simulated process that starts with person A being infected**

- Hellewell, Joel, et al. "Feasibility of controlling COVID-19 outbreaks by isolation of cases and contacts." The Lancet Global Health (2020):e488-e496.

This page is not lectured.

Branching Process

- Consider a population consisting of individuals able to produce offspring of the same kind. Suppose that each individual will, by the end of its lifetime, have produced j new offspring with probability P_j, $j \geq 0$, independently of the numbers produced by other individuals.
- Let X_n be the size of the n-th generation.
- Let μ and σ^2 be the mean and variance of one individual's offsprings, i.e.

$$\mu = \sum_{j=0}^{\infty} jP_j \text{ and } \sigma^2 = \sum_{j=0}^{\infty} (j-\mu)^2 P_j.$$

- For given X_0, let us compute $E[X_n]$ and $Var(X_n)$.

Branching Process

- **Proposition:** If $P_0 > 0$, the population will either die out or explode (its size converges to infinity) . Let $\pi_0 = \lim_{n\to\infty} P(X_n = 0|X_0 = 1)$. Then $\pi_0 = 1$ if $\mu \leq 1$, otherwise π_0 satisfies

$$\pi_0 = \sum_{j=1}^{\infty} \pi_0^j P_j.$$

- **Example 4.32** If $P_0 = 1/2, P_1 = 1/4, P_2 = 1/4$, then determine π_0
- **Example 4.33** If $P_0 = 1/4, P_1 = 1/4, P_2 = 1/2$, then determine π_0.

Markov Chain Monte Carlo

- Suppose X is a random vector taking value in $\{x_j\}_{j\geq 1}$ and we wish to compute $E[h(X)]$.
- Monte Carlo simulation is a numerical algorithm that use random numbers to generate a partial sequence of i.i.d. copies of X, i.e. $X_1, X_2, ..., X_n$. By the strong law of large numbers,

$$\lim_{n\to\infty} \frac{\sum_{i=1}^n h(X_i)}{n} = E[h(X)].$$

- Note that if X_t is an irreducible ergodic Markov chain, we have

$$\lim_{n\to\infty} \frac{\sum_{i=1}^n h(X_i)}{n} = E_\pi[h(X)].$$

- A Monte Carlo simulation that generates Markov chain instead of i.i.d. sequence is called a Markov chain Monte Carlo (MCMC).

irreducible + positive recurrent $\Longrightarrow$

ergodic { irreducible, positive recurrent, aperiodic }

Markov Chain Monte Carlo

- MCMC is widely used in Bayesian statistics.

- Suppose we need to estimate $E[h(X)]$ where we only know that

$$P(X = x_j) = \frac{b(j)}{B}, \text{ where } B \text{ is unknown.}$$

  MCMC can simulate X without knowing B by constructing a special Markov chain whose stationary distribution $\pi(j) = b(j)/B$.

- Main idea of MCMC: we construct a Markov chain with stationary distribution $\pi(j) = b(j)/B$ without knowing the value of B.

Markov Chain Monte Carlo

- **Example: (Hastings-Metropolis algorithm)** Let $Q = (q_{ij})$ be the transition probabilities of any irreducible Markov chain on $\mathcal{S} = \{1, 2, ...\}$. Define a Markov chain X_t with transition probabilities

$$p_{ij} = \begin{cases} q_{ij}\alpha_{ij}, & \text{for } i \neq j, \\ q_{ii} + \sum_k q_{ik}(1 - \alpha_{ik}), & i = j \end{cases}$$

with

$$\alpha_{ij} = \min\left(\frac{b(j)q_{ji}}{b(i)q_{ij}}, 1\right).$$

Then, the stationary distribution of X_t is $\pi(j) = b(j)/B$.

Practice. We now generate chain X_t.

- when $X_n = i$, generate $Y = j$ with probab. q_{ij}.
- If $Y = j$, set

$$X_{n+1} = \begin{cases} j, & \text{with probab. } \alpha_{ij}; \\ i, & \text{" " } 1 - \alpha_{ij}. \end{cases}$$

Thm. Assume the chain X_t is recurrent. Then

$$\lim_{n\to\infty} \frac{\sum_{i=1}^{n} h(X_i)}{n} = E_\pi[h(X)].$$

for any fun $h(x)$ s.t. $E_\pi|h(x)| < \infty$.

Proof. We need to show $\{p_{ij} : i,j \in S\}$ is a transition probability matrix and

$$\pi(i)p_{ij} = \pi(j)p_{ji}.$$

- From def,

$$p_{ii} = q_{ii} + \sum_{k\neq i} q_{ik}(1-\alpha_{ik})$$

$$\overset{\alpha_{ii}=1}{=} \sum_k q_{ik} - \sum_{k\neq i} q_{ik}\alpha_{ik}$$

$$= 1 - \sum_{k\neq i} q_{ik}\alpha_{ik}.$$

Thus, $\sum_j p_{ij} = 1$ for any i.

- By def.

$$\alpha_{ij} = \min\left(\frac{b_j q_{ji}}{b_i q_{ij}}, 1\right).$$

Then, for $i \neq j$,

$$\pi(i)p_{ij} = \pi(i)q_{ij}\alpha_{ij} = \pi(i)q_{ij} \cdot \min\left(\frac{\pi(j)q_{ji}}{\pi(i)q_{ij}}, 1\right)$$

$$= \min(\pi(i)q_{ij}, \pi(j)q_{ji})$$

$$= \pi(j)p_{ji}.$$

Eg. Let $\varphi = \{(x_1, \cdots, x_m) : \text{a permutation of } \{1, 2, \cdots, m\} \text{ with } \sum_{j=1}^{m} j x_j > a\}$

where a is given. Please sample a point $(x_1, \cdots, x_m)$ uniformly over φ.

Solution. Now $\pi(i) = \frac{1}{|\varphi|}$, $i \in \varphi$. Headache: no close form for $|\varphi|$.

We need to construct a MC $q(i,j)$ first.

Neighbor: we say $(x_1, \cdots, x_m)$ and $(y_1, \cdots, y_m)$ are neighbors if the latter is obtained by swapping x_i and x_j for $i \neq j$.

e.g., $(1,2,3,4)$ and $(1,2,4,3)$ are neighbors.

$(1,2,3,4)$ and $(1,3,4,2)$ are not.

Define a MC/random walk with transition probab.

$$q(s,t) = \frac{1}{|N(s)|}, \text{ if } t \text{ is a neighbor of } s,$$

where $N(s)$ is the neighbor of s, and $|N(s)| = \#$ of neighbors of s.

This is a recurrent MC. Now

$$\alpha(s,t) = \min\left\{\frac{q(t,s)\,\pi(t)}{q(s,t)\,\pi(s)}, 1\right\}$$

$$= \min\left\{\frac{|N(s)|}{|N(t)|}, 1\right\}.$$

Then follow the receipe. Note that the next step the chain visits is always a neighbor of the current state.

Special case of Hastings-Metropolis algorithm: Gibbs Sampler

Story: $X = (X_1, \cdots, X_n)$ where each X_i is discrete

Purpose: Generate a random vector with probab. mass fun

$$p(x) = c\, g(x)$$

where $g(x)$ is known but c is not.

Procedure. Given the present state $x = (x_1, \cdots, x_n)$, the chain goes to another state in the following way:

- randomly choose a number in $\{1, 2, \cdots, n\}$.
- If i is chosen, compute the conditional pmf of X_i given the remaing ones:

$$p(X = x) = p(X_i = x \mid X_j = x_j, j \neq i)\, \frac{1}{n}$$

$$= \frac{p(y)}{n\, p(X_j = x_j, j \neq i)} = \frac{g(x_1, \cdots, x_{i-1}, x, x_{i+1}, \cdots, x_n)}{n \sum_x g(x_1, \cdots, x_{i-1}, x, x_{i+1}, \cdots, x_n)}$$

- Sample point x and the chain moves to $y = (x_1, \cdots, x_{i-1}\ x, x_{i+1}, \cdots, x_n)$.

Justification. In this case we pick

$$q(x,y) = \frac{1}{n} p(X_i = x \mid X_j = x_j,\ j \neq i).$$

Now

$$\alpha(x,y) = \min\left\{ \frac{p(y)\,q(y,x)}{p(x)\,q(x,y)},\ 1 \right\}$$

$$= \min\left\{ \frac{p(y)\,p(x)}{p(x)\,p(y)},\ 1 \right\}$$ ← Common denominators in $q(x,y)$ and $q(y,x)$

$$= 1.$$

Thus, the candidate/new state is always accepted in the Hastings-Metropolis algorithm.

Cont. case also holds!

Ex 4.43. $X_1, \cdots, X_n$ indept, $X_i \sim \text{Exp}(\lambda_i)$. $S = \sum_1^n X_i$.

Purpose: generate $\mathbf{X} = (X_1, \cdots, X_n)$ given $S > c$, where $c > 0$ is a const. Then $\mathbf{X}$ has pdf

$$f(x_1, \cdots, x_n) = \frac{1}{p(S>c)} \prod_{i=1}^n \lambda_i e^{-\lambda_i x_i},\quad x_i \geq 0,\ \sum_1^n x_i > c.$$

Procedure.

- Starting an initial vector $\mathbf{x} = (x_1, \cdots, x_n)$ with $x_i \geq 0$ for each i and $\sum_1^n x_i > c$.
- Generate I uniformly in $\{1, 2, \cdots, n\}$.

- Generate X with pdf

$$f(x) = \frac{f(x_1, \ldots, x_I, \ldots x_n)}{\int f(x_1, \ldots, x_I, \ldots, x_n)\, dx_I}$$

$$= \lambda_I e^{-\lambda_I x_I}, \quad x_I + \sum_{i \neq I} x_i > c.$$

In other words, generate $X \sim Exp(\lambda_I)$ given $x_I > c - \sum_{i \neq I} x_i$. By the memoryless property,

$$P(U > s+t \mid U > s) = P(U > t)$$

if $U \sim Exp(\lambda)$. Thus first generate $Y \sim Exp(\lambda_I)$. Then set $X = Y + \left(c - \sum_{i \neq I} x_i\right)^+$, where

$x^+ = \max\{x, 0\}$

- The value of $x_I \to X$ and

$(x_1, \ldots, x_n) \to (x_1, \ldots, x_{I-1}, X, x_{I+1}, \ldots, x_n)$

- **Proposition 4.5** If i is positive recurrent and communicates with j, then j is positive recurrent.

Lemma 1. Let X_0, X_n be r.v.'s s.t. $X_n \to X_0$ in probab, i.e., $\lim_n P(|X_n - X_0| \geq \varepsilon) = 0$ for any $\varepsilon > 0$. Assume $|X_n| \leq a$ for all $n \geq 0$ where a is a const not depending on n. Then $\lim_{n\to\infty} EX_n = EX_0$. The conclusion also holds if $X_n \to X_0$ a.s.

Dominated convergence thm

Proof. Notice $|EX_n - EX_0| \leq E|X_n - X_0|$. Set $A = \{|X_n - X_0| \geq \varepsilon\}$. Then

$$E|X_n - X_0| = E(|X_n - X_0| I_A) + E(|X_n - X_0| I_{A^c})$$
$$\leq 2a \cdot P(|X_n - X_0| \geq \varepsilon) + \varepsilon$$

for any $\varepsilon > 0$. Let $n \to \infty$, then let $\varepsilon \to 0$ we get $E|X_n - X_0| \to 0$. Since $X_n \to X_0$ a.s. implies $X_n \to X_0$ in probab, the 2nd conclusion holds.

Proof of Prop 4.5. By assumption, $\pi_i > 0$ and $p_{ij}^m > 0$ for some $m \geq 1$. By def.

$$\pi_j = \lim_{T\to\infty} \frac{1}{T} \sum_{n=1}^{T} I(X_n = j)$$

Notice $0 \leq \frac{1}{T}\sum_{n=1}^{T} I(X_n = j) \leq 1$. By Lemma 1,

$$\pi_j = \lim_T \frac{1}{T} \sum_{n=1}^{T} P(X_n = j) \geq \lim_T \frac{1}{T} \sum_{n=m}^{T} P(X_n = j \mid X_{n-m} = i) P(X_{n-m} = i)$$
$$= \lim_T \frac{1}{T} \sum_{n=m}^{T} p_{ij}^m \cdot P(X_{n-m} = i) = p_{ij}^m \cdot \pi_i > 0.$$

Since $\lim_T \frac{1}{T} \sum_{n=m}^{T} P(X_{n-m} = i) \overset{k=n-m}{=} \lim_{T\to\infty} \frac{T-m}{T} \cdot \frac{1}{T-m} \sum_{k=0}^{T-m} P(X_k = i) = \pi_i$.

Chapter Summary

- Discrete time Markov chain: definition, Markov property. transition probab. matrix.
- Chapman-Kolmogorov equations: computing transition probabilities
- Long-run behavior of DTMC
  - classification of states
  - transient, recurrent, positive/null recurrent
  - irreducible
  - aperiodic, ergodic
  - long-run proportion, stationary probability, limiting probability...
- Transient behavior of DTMC
- Time reverse, reversible MC
- Application examples: branching process, MCMC (Hastings-Metropolis, Gibbs sampler)
  Gambler ruin problem